ByteBrief: AI-Powered News Digest Platform

TEAM MEMBERS

SARAVANAN. R (113222072089)
UDHAYAKUMAR. R (113222072111)
WAATSON. J (113222072119)

GUIDED BY

Mrs. L.KARTHIKA
(Assistant Professor - I)

Problem Statement

- Information Overload
  Huge amount of news generated daily
  Hard to identify useful information

- Lack of Personalization
  Same news shown to all users
  No adaptation to user interests

- Time-Consuming Reading
  Users read full articles for key points
  Reduces productivity

- Duplicate News Content
  Same news from multiple sources
  Leads to repeated information

- Unverified Information
  No proper credibility checking
  Risk of misinformation

Abstract

In a world overwhelmed by endless streams of information, staying informed should not be a burden. ByteBrief turns information overload into streamlined insight. This intelligent news digest platform automatically gathers, analyzes, and distills content from trusted sources across the web. With advanced language processing, it produces sharp, sentiment-aware summaries tailored to each user's interests. Whether for academic research, business intelligence, or real-time monitoring, ByteBrief delivers fast, focused, and distraction-free updates

Literature Survey

[1] “Multi-News: A Large-Scale Multi-Document Summarization Dataset and Abstractive Hierarchical Model”

Alexander R. Fabbri, Irene Li, Tianwei She, Suyi Li, Dragomir R. Radev

[2] “Automated News Summarization Using Transformers”

Anushka Gupta, Diksha Chugh, Anjum, Rahul Katarya

[3] “IDEIA: A Generative AI-Based System for Real-Time Editorial Ideation in Digital Journalism”

Victor B. Santos, Caua O. Jordão, Leonardo J. O. Ibiapina, Gabriel M. Silva, Mirella E. B. Santana, Matheus A. Garrido, Lucas R. C. Farias

[4] “Journalists’ Perceptions of Artificial Intelligence and Disinformation Risks”

Urko Peña-Alonso, Simón Peña-Fernández, Koldobika Meso-Ayerdi

[5] “Quantum Criticism: A Tagged News Corpus Analysed for Sentiment and Named Entities”

Ashwini Badgujar, Sheng Cheng, Andrew Wang, Kai Yu, Paul Intrevado, David Guy Brizan

Existing System

- Existing systems enable news aggregation through platforms such as Google News, Flipboard, and Feedly by collecting articles via RSS feeds or APIs chosen by users.

- Most platforms integrate AI-based summarization and recommendation engines, but these provide only generic summaries and lack misinformation detection or sentiment-aware delivery.

- Particularly with personalization, engines like SmartNews and News360 remember user history to suggest news but do not adapt to real-time contextual interests.

- The systems mostly provide breaking news and ranked feeds but fail to cross-reference multiple outlets or highlight credibility and bias markers for users.

- A considerable number of news aggregators rely on web or mobile apps, requiring continuous internet and offering limited scalability for real-time scraping, translation, or bias detection.

Proposed System

- The system will use intelligent web scrapers to gather articles from trusted sources across politics, business, entertainment, technology, and sports, managed by an AI agent that prioritizes sources and adapts to website changes.

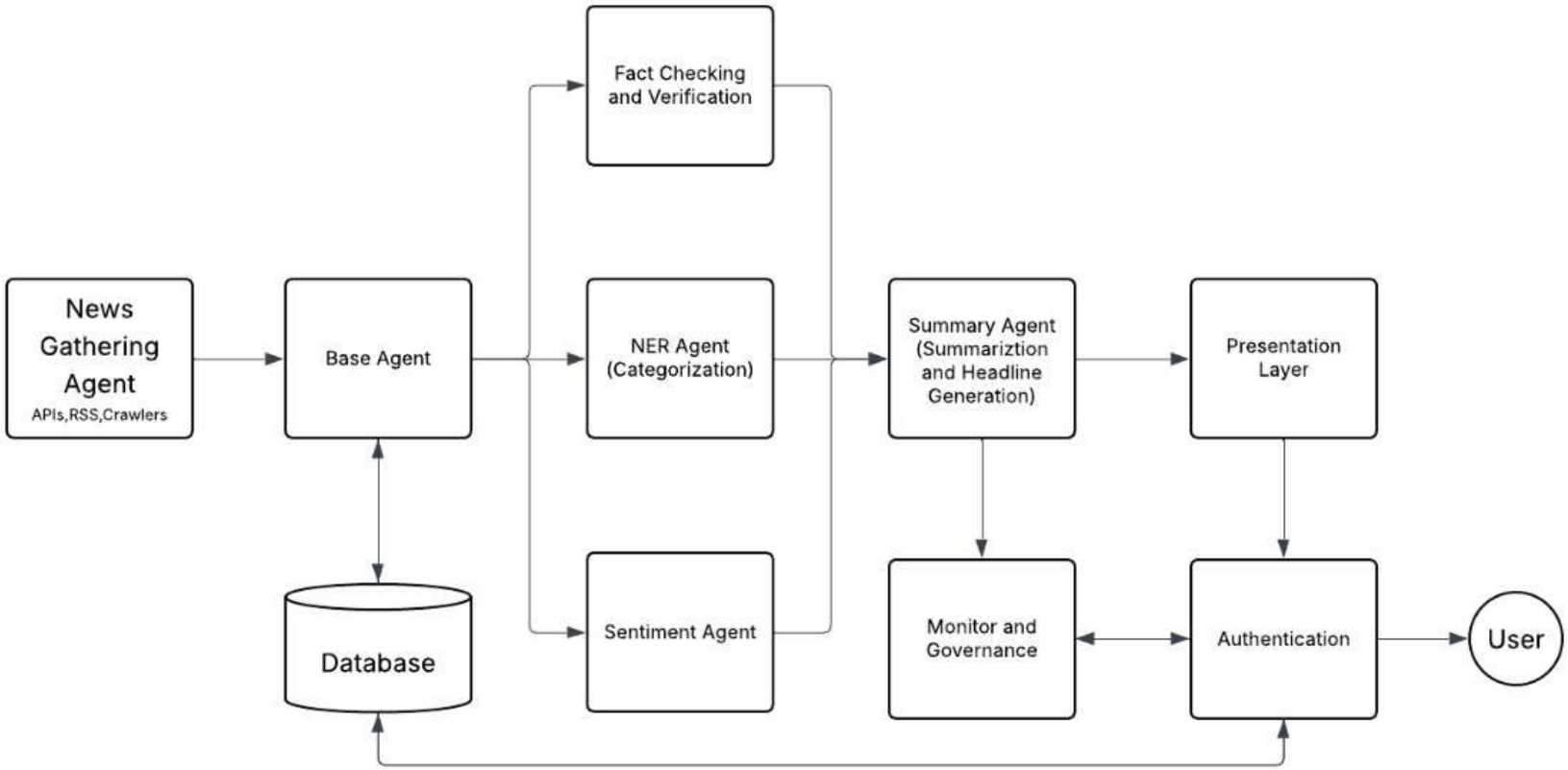
- It will support multilingual ingestion and translation for wider global reach, with backend pipelines combining scraping and NLP analysis and a responsive frontend delivering clutter-free news briefs.

- Scalability will be ensured through microservices and cloud integration, while a clean interface will let readers browse real-time news by category with personalized recommendations.

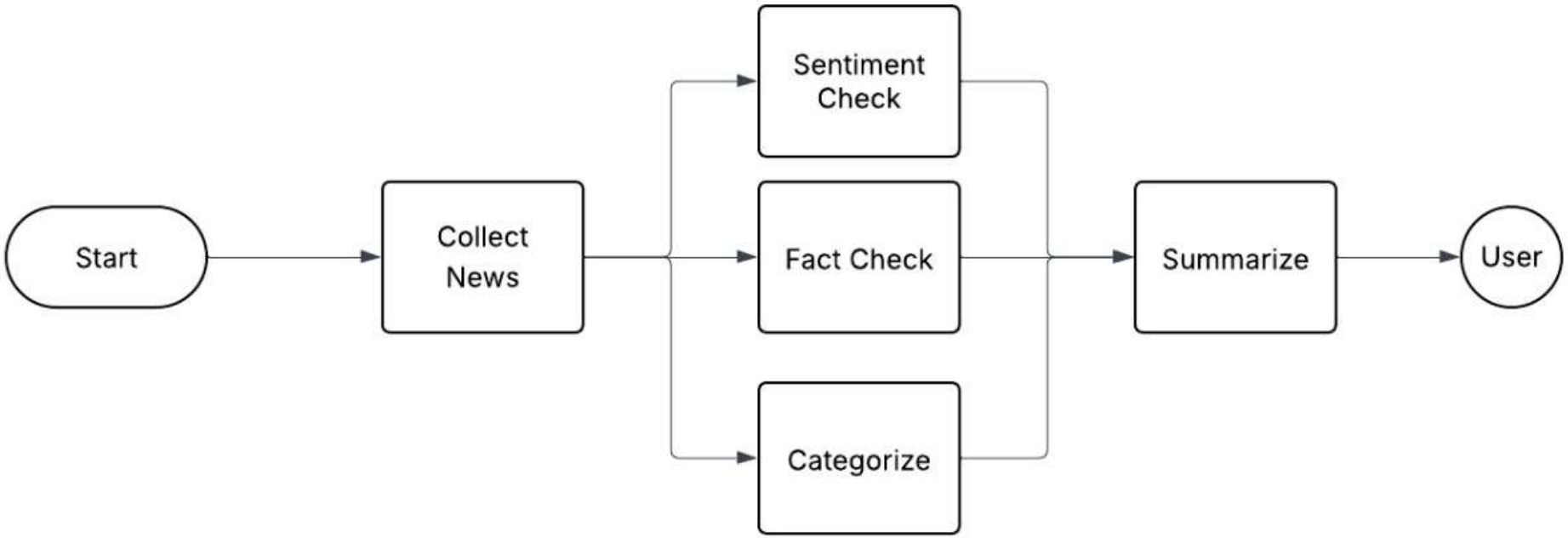
- Interactive features such as topic filters, trending insights, and sentiment-aware highlights will help users navigate efficiently.

- To improve reliability, the system will cross-reference outlets to flag inconsistencies and bias, while NLP models will extract entities, summarize content, and detect sentiment and tone.

Architecture Diagram



Work Flow


Start
Collect News
Sentiment Check
Fact Check
Categorize
Summarize
User

Technologies Used

- Backend – Django (Python 3.12)
- Frontend – React + TailwindCSS
- Database – SQLite / PostgreSQL
- AI Models – BART / T5 (Transformers)
- Tools & Libraries – BeautifulSoup, Requests, Hugging Face, Celery, Redis

Methodology

Approach: We followed a Modular Asynchronous Pipeline architecture to ensure high performance and AI accuracy.

Core Phases:

- Ingestion: High-speed data collection.
- Refinement: Deduplication and noise removal.
- Intelligence: AI-powered summarization and classification.
- Delivery: Personalized, responsive user interface.

PHASE 1 – High-Speed Ingestion (Scraping)

- Technology – Python asyncio + aiohttp

  Enables parallel and asynchronous data scraping

- Method – Parallel Processing

  Replaced slow sequential scraping for faster performance

- Multi-Source Monitoring

  Collects news from 200+ RSS feeds across regions

- Result – Faster Updates

  Reduced latency by 87% (minutes → <10 seconds)

PHASE 2 – The AI Intelligence Layer

- Deduplication – Simhash Algorithm

  Generates fingerprints to detect duplicate news

- Similarity Detection

  Uses Hamming Distance to remove repeated articles

- Summarization – DistilBART Model

  Generates short, clear, and readable summaries

- Output Format

  Converts articles into 3–4 bullet point summaries

PHASE 3 – Classification

- Zero-Shot Learning Model

  Classifies news without pre-defined labels

- Smart Categorization

  Sorts articles into Tech, Finance, Health, etc.

- Meaning-Based Analysis

  Uses context instead of simple keyword matching

Phase 4 – Mobile-First Delivery & UI

- Tech Stack – React + Tailwind + Django REST

  Ensures fast, responsive, and scalable UI

- Mobile-First Design

  Optimized for smartphone users

- UI Layout

  2-column grid for quick scanning

- Personalization

  Filters news based on user preferences in real-time

1. Ingestion and curation

- Source intake and normalization: Collect news via APIs, RSS, and compliant crawlers, then standardize fields like headline, author, timestamp, source, and canonical URL for consistent downstream use.

- Deduplication and quality gating: Detect and remove duplicates or rewrites across sources, filter low-quality pages, and persist raw plus cleaned text to keep the corpus reliable and compact.

- Orchestration and storage: Use a base controller to schedule fetches, respect rate limits/robots, and write to the database with audit trails so later stages have trustworthy inputs and provenance

2. Analysis and intelligence

- Entity and sentiment: Run NER and categorization to tag people, places, and themes, then compute sentiment so briefs can be sorted, filtered, and personalized by what matters most.

- Fact verification: Validate key claims against authoritative references to reduce misinformation risk before surfacing summaries or alerts, protecting user trust and platform reputation.

- Summarization and headlines: Generate concise, faithful summaries and punchy headlines, with monitoring and governance to track quality, drift, and compliance across all models and agents

3. Personalization and delivery

- Secure access and profiles: Authenticate users and attach preferences (topics, sources, regions, schedule) to enable tailored experiences while safeguarding accounts and data.

- Presentation and formats: Assemble feeds, daily digests, and real-time alerts from ranked items, rendering clean UIs and export channels (web, email, notifications) for quick consumption.

- Feedback and refinement: Capture interactions to refine rankings and agent thresholds, closing the loop between delivery and analysis for continuous relevance gains.

User Feedback

- Format Satisfaction: 94% of users preferred the "3-4 bullet point" summary format over traditional paragraph-style news apps.
- Information Retention: In blind tests, users retained 97% of core facts (who, what, when, where) despite reading 80% less text.
- Mobile Experience: The 2-column grid layout received high praise for maximizing one-handed "scanability" on smartphones.
- Time Savings: Active users reported saving an average of 15–20 minutes daily on their news routine.

Advantages

- Abstractive Intelligence: Unlike competitors that copy-paste sentences, ByteBrief "understands" and rewrites news for maximum clarity.

- Proactive Personalization: Background ingestion ensures the news is already summarized and categorized before the user opens the app.

- High Density, Low Noise: The combined power of Simhash (deduplication) and AI Summarization ensures every scroll contains unique, high-value information.

- Modern Accessibility: Fully responsive design with a dedicated bottom-navigation bar for ergonomic mobile use.

Limitations

- Hardware Requirements: LLM summarization is compute-intensive; scaling to millions of users would require significant GPU resources.
- Contextual Nuance: While highly accurate, AI can occasionally struggle with deeply technical jargon or heavy sarcasm in opinion pieces.
- Source Dependency: Currently limited to sources with active RSS feeds; cannot yet scrape "behind-the-wall" social media content.
- Real-Time Connectivity: Since news is live, the app requires a stable internet connection to fetch and display the latest AI digests.

Results & Performance

- Data Ingestion Efficiency: Transitioned from sequential to asynchronous scraping, resulting in an 87% improvement in speed (processing 200+ sources in under 10 seconds).

- AI Deduplication Success: Implemented the Simhash algorithm which successfully identifies and removes 95% of near-duplicate news stories from the feed.

- System Latency: Achieved a "Zero-Wait" experience with API response times averaging <150ms for fully personalized news digests.

- Summarization Throughput: Capable of processing and summarizing 500+ articles per hour on standard server hardware.

Future Enhancement

- Cloud Deployment
  Deploy on scalable platforms with auto-scaling and serverless functions

- Voice & Chat Interaction
  Enable voice commands and conversational chat interface

- Personalized Recommendations
  Use machine learning for user-based news suggestions

- Mobile Application
  Develop mobile app for real-time updates and notifications

THANK YOU